



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

Summary of Methodologies

1. Data Collection

- Collected Falcon 9 launch data using the SpaceX REST API.
- Gathered additional launch information by scraping Wikipedia using BeautifulSoup.

2. Data Preparation

- Cleaned and formatted the collected datasets.
- Performed exploratory data analysis and SQL queries to better understand the launch data.

3. Visualization

- Created visualizations to explore relationships between launch variables.
- Built an interactive Folium map and a Plotly Dash dashboard.

4. Machine Learning

- Trained and evaluated multiple classification models.
- Used GridSearchCV to optimize model performance.

Summary of Results

Data Insights

Launch success rates improved over time. Launch site, payload mass, and orbit showed noticeable relationships with landing success.

Model Performance

Decision Tree achieved the highest test accuracy of **94.44%**.

Logistic Regression, SVM, and KNN also produced strong predictive performance.

Conclusion

Historical launch data can be used to successfully predict Falcon 9 first-stage landing outcomes. The Decision Tree model provided the best overall prediction accuracy.

Introduction

Project Background and Context

- The goal of this project is to predict whether the first stage of a SpaceX Falcon 9 rocket will successfully land after launch. Since SpaceX reuses its boosters, successful landings help reduce launch costs and make launches more affordable than many competitors. Predicting landing success can provide useful insights into mission performance and estimated launch costs.

Questions Addressed

- Which factors are most closely related to successful Falcon 9 first-stage landings?
- Can historical launch data be used to accurately predict landing outcomes?
- Which machine learning model provides the best prediction performance?

Section 1

Methodology

Methodology

- **Data Collection**
 - Collected Falcon 9 launch data from the SpaceX API.
 - Retrieved additional launch records from Wikipedia using web scraping.
- **Data Preparation**
 - Cleaned and organized the collected data.
 - Prepared the dataset for analysis and model training.
- **Exploratory Data Analysis**
 - Explored launch trends using visualizations and SQL queries.
 - Examined relationships between launch outcomes and key variables.
- **Interactive Visualizations**
 - Built an interactive launch site map using Folium.
 - Developed a Plotly Dash dashboard for interactive data exploration.
- **Predictive Modeling**
 - Trained and compared multiple classification models.
 - Tuned model parameters and evaluated prediction performance.

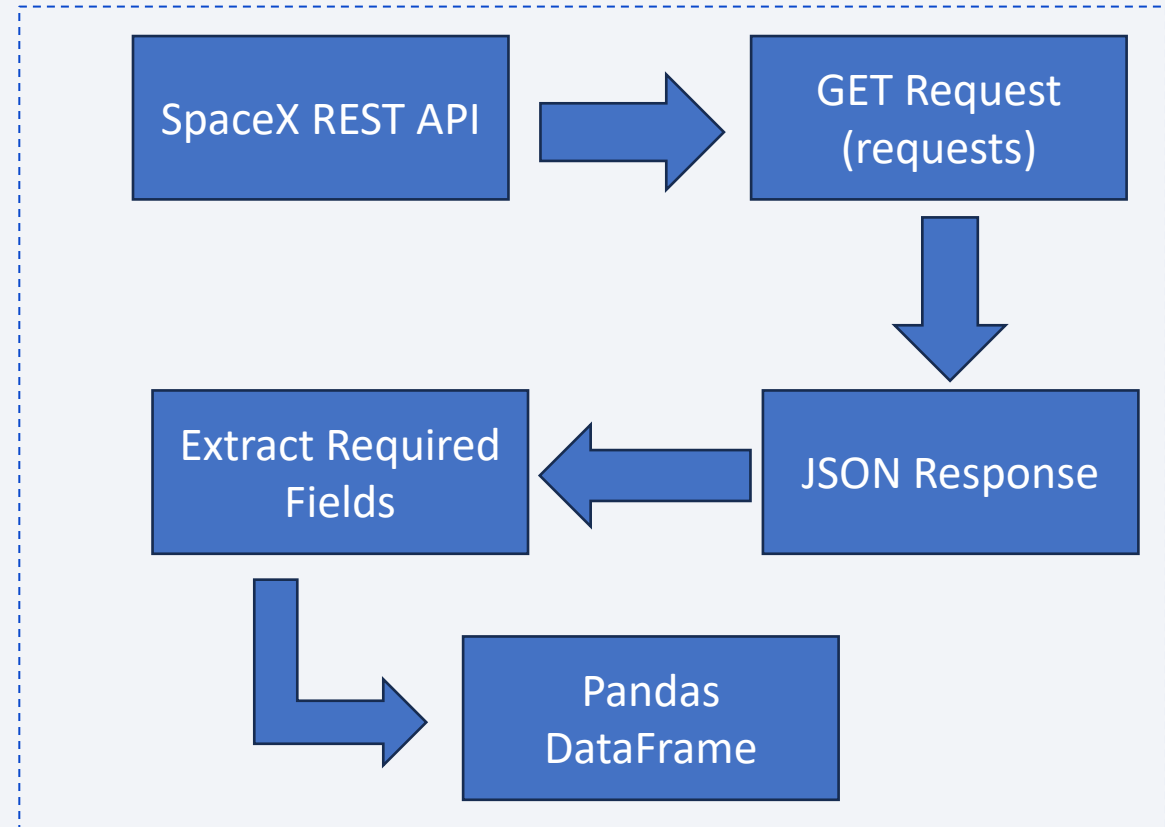
Data Collection

- **API Data Collection**
- Retrieved Falcon 9 launch data using the SpaceX REST API.
- Extracted launch details and converted the response into a Pandas DataFrame.
- **Web Scraping**
- Collected additional launch records from Wikipedia using BeautifulSoup.
- Parsed the HTML tables and stored the data in a Pandas DataFrame.

Data Collection – SpaceX API

SpaceX API Data Collection

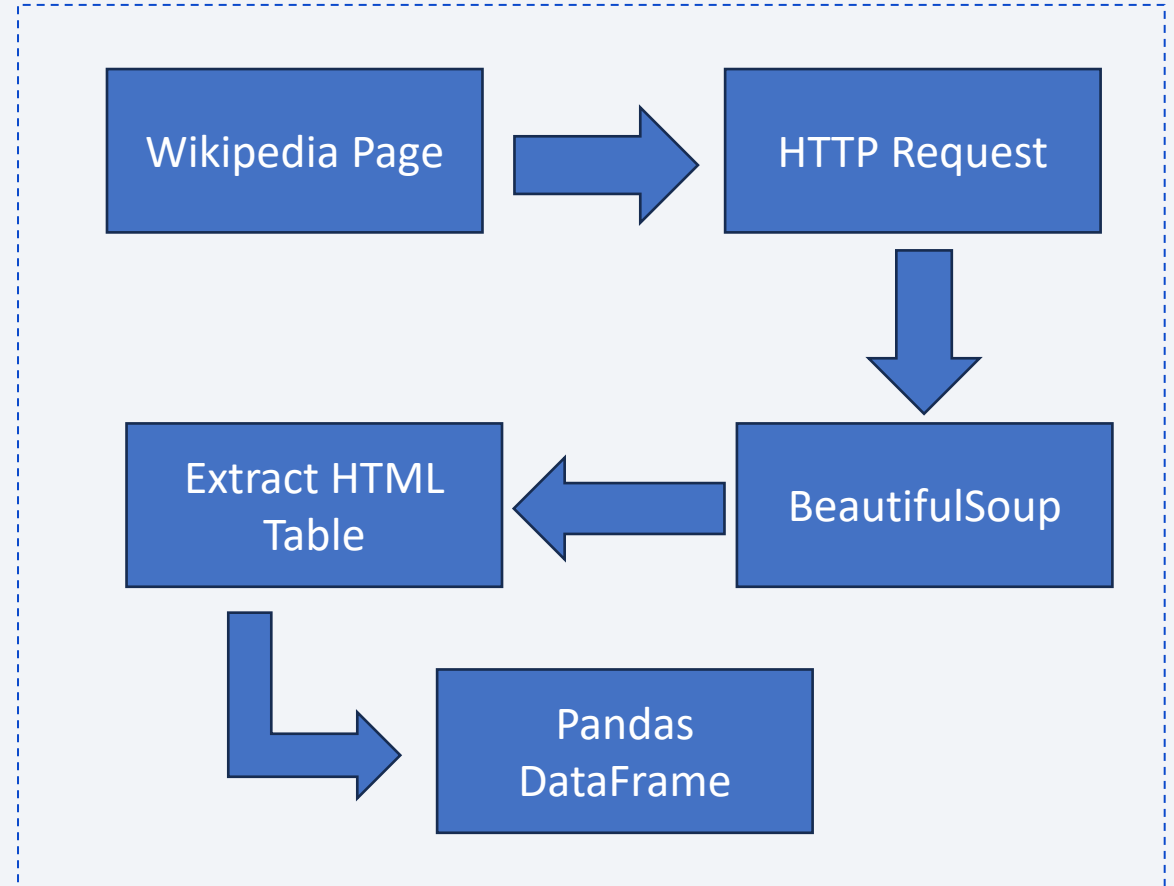
- Retrieved Falcon 9 launch data using the SpaceX REST API.
- Sent GET requests with the requests library.
- Extracted launch, payload, booster, and launch site information.
- Converted the API response into a Pandas DataFrame for analysis.



Data Collection - Scraping

Web Scraping Process

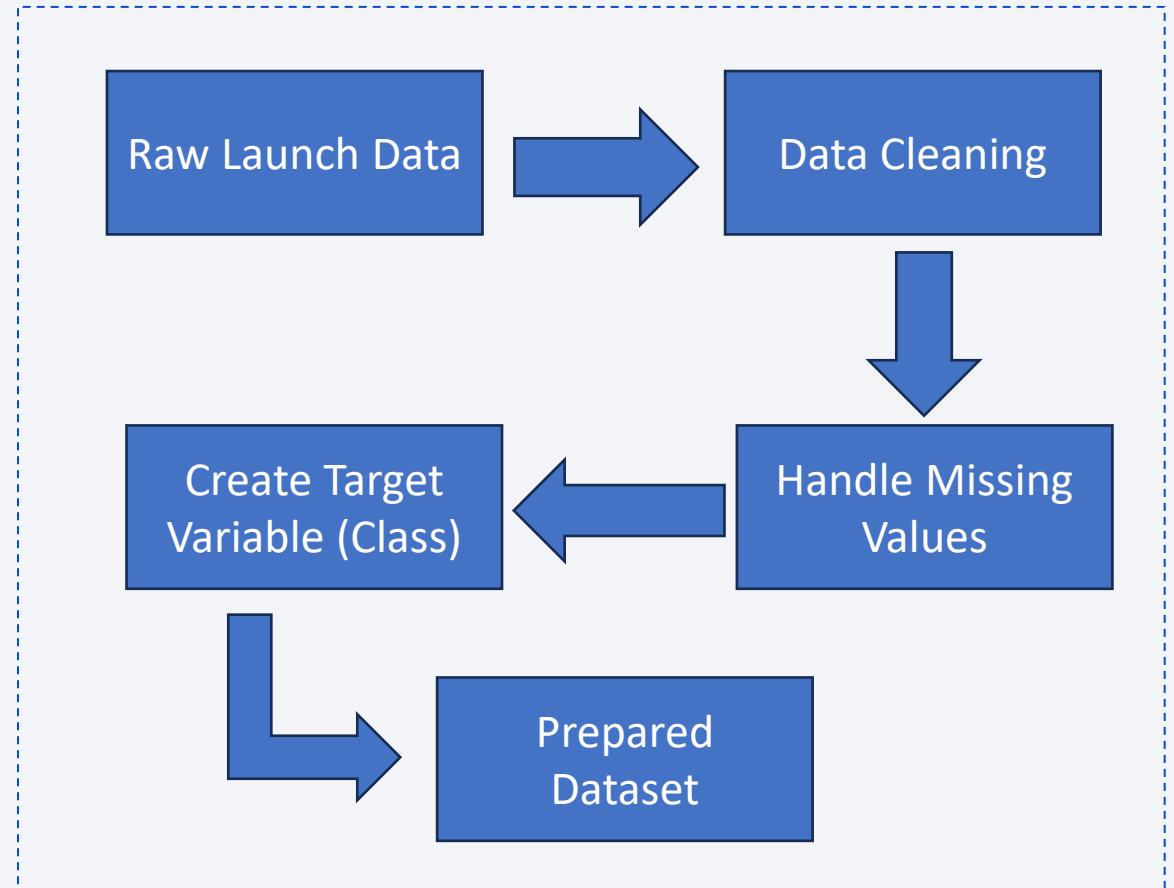
- Retrieved Falcon 9 launch records from a Wikipedia webpage.
- Parsed the HTML content using BeautifulSoup.
- Extracted launch information from the table.
- Stored the scraped data in a Pandas DataFrame for analysis.



Data Wrangling

Data Preparation

- Loaded the collected launch dataset into a Pandas DataFrame.
- Identified and handled missing values.
- Filtered the data to include Falcon 9 launches only.
- Converted landing outcomes into a binary classification variable (Class) for machine learning.



EDA with Data Visualization

Exploratory Analysis

- Created scatter plots to examine the relationship between payload mass and launch success.
- Compared launch outcomes across different launch sites and orbit types.
- Analyzed launch success trends over time.
- Applied one-hot encoding to categorical variables to prepare the dataset for machine learning.

Purpose

- Better understand the launch data.
- Identify variables related to landing success.
- Support feature selection for predictive modeling.

EDA with SQL

SQL Analysis Process

- Loaded the launch dataset into a SQLite database for structured querying.
- Queried launch records based on launch sites, booster versions, payloads, and landing outcomes.
- Used SQL aggregation functions such as COUNT(), AVG(), MAX(), and SUM() to summarize launch statistics.
- Filtered records using WHERE clauses to analyze specific launch conditions.
- Sorted and ranked results with ORDER BY to identify trends over time.
- Grouped launch records using GROUP BY to compare launch outcomes across different categories.

Purpose of SQL Analysis

- Explore the launch dataset from multiple perspectives.
- Identify patterns in payloads, launch sites, and landing outcomes.
- Validate insights discovered during the visualization stage.
- Support the selection of meaningful features for predictive modeling.

Build an Interactive Map with Folium

Map Development Process

- Created an interactive map centered around SpaceX launch locations using Folium.
- Added markers to display the locations of each launch site.
- Used marker clusters to organize individual launch records and improve map readability.
- Colored markers to distinguish successful and unsuccessful launches.
- Drew lines and calculated distances between launch sites and nearby landmarks using the Haversine formula.

Purpose of the Interactive Map

- Visualize the geographic distribution of SpaceX launch sites.
- Compare landing outcomes across different launch locations.
- Analyze how nearby geographic features may relate to launch operations.

Build a Dashboard with Plotly Dash

Dashboard Development

- Built an interactive dashboard using Plotly Dash to visualize Falcon 9 launch data.
- Added a launch site dropdown menu to filter results by location.
- Included a payload range slider to analyze launches within different payload ranges.
- Created interactive pie charts to compare successful launches across launch sites.
- Developed scatter plots to explore the relationship between payload mass and landing success.

Purpose of the Dashboard

- Enable interactive exploration of launch data.
- Compare launch performance across different launch sites.
- Examine how payload mass relates to landing outcomes.
- Provide an intuitive way to analyze launch trends.

Predictive Analysis (Classification)

Model Development

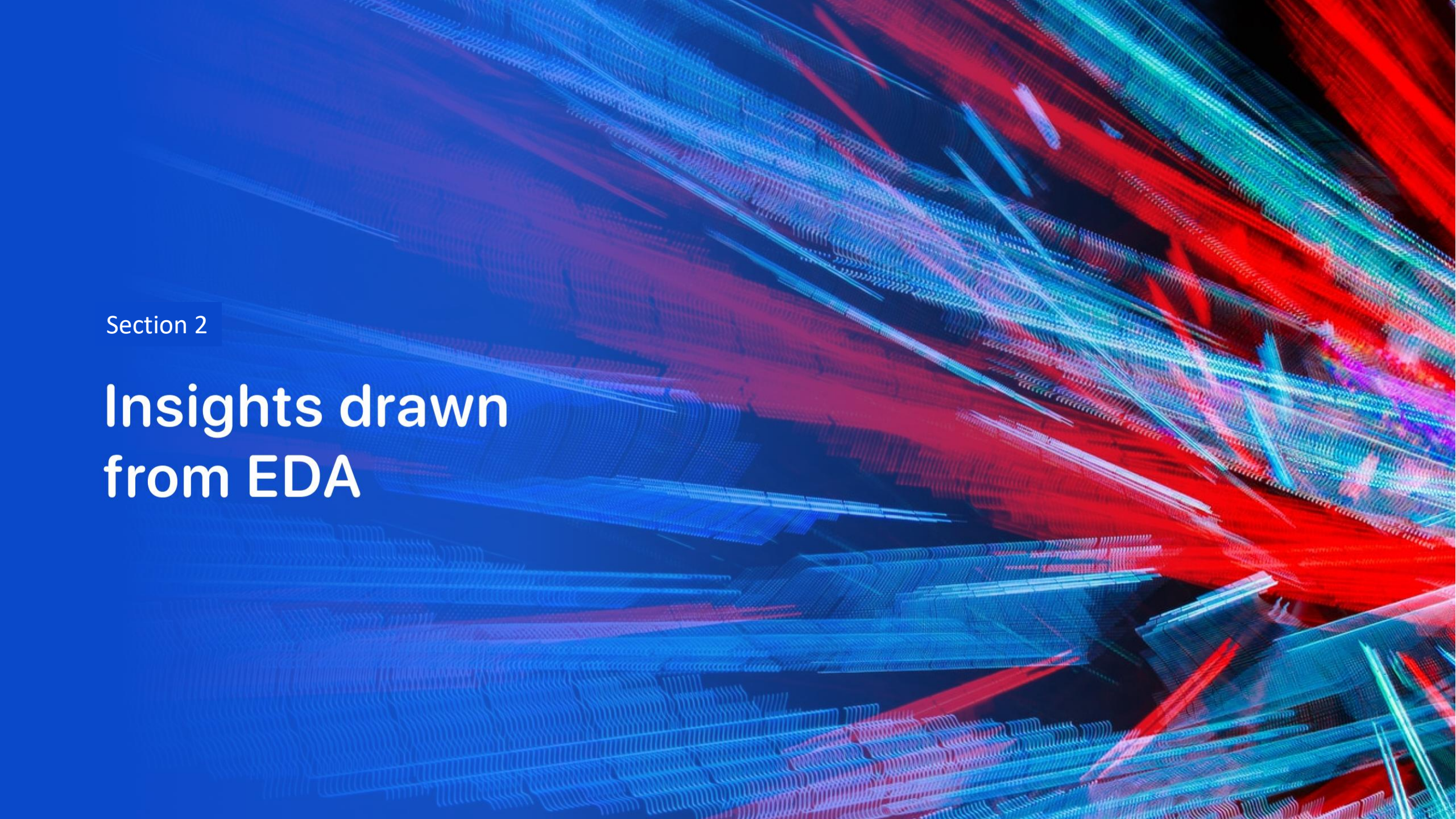
- Selected relevant features from the processed dataset and defined the target variable (Class).
- Standardized the feature data and split it into training and testing sets.
- Trained multiple classification models, including Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- Used GridSearchCV with cross-validation to optimize model hyperparameters.
- Evaluated each model using test accuracy and confusion matrices to identify the best-performing classifier.

Purpose of Predictive Analysis

- Build a model capable of predicting Falcon 9 first-stage landing success.
- Compare the performance of different classification algorithms.
- Select the model with the highest predictive accuracy.

Results

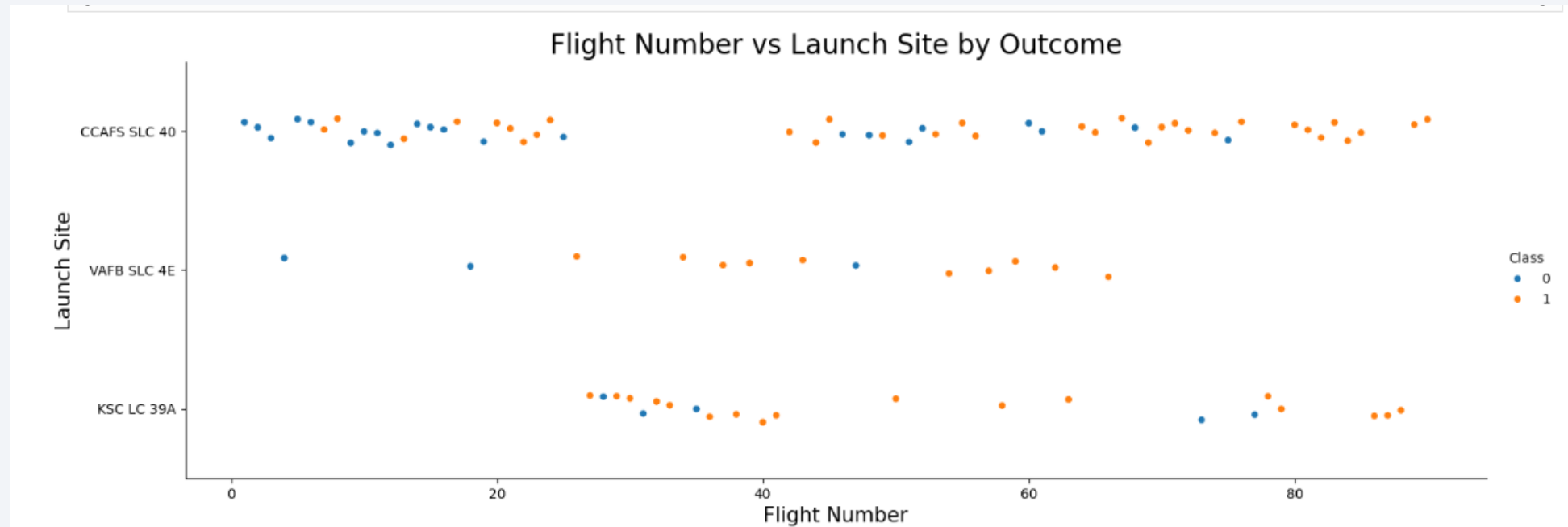
- The following sections present the findings from each stage of the project:
- **Exploratory Data Analysis**
 - Data visualization results
 - SQL query results
- **Interactive Analytics**
 - Folium interactive map
 - Plotly Dash dashboard
- **Predictive Analysis**
 - Classification model comparison
 - Best-performing model and evaluation



Section 2

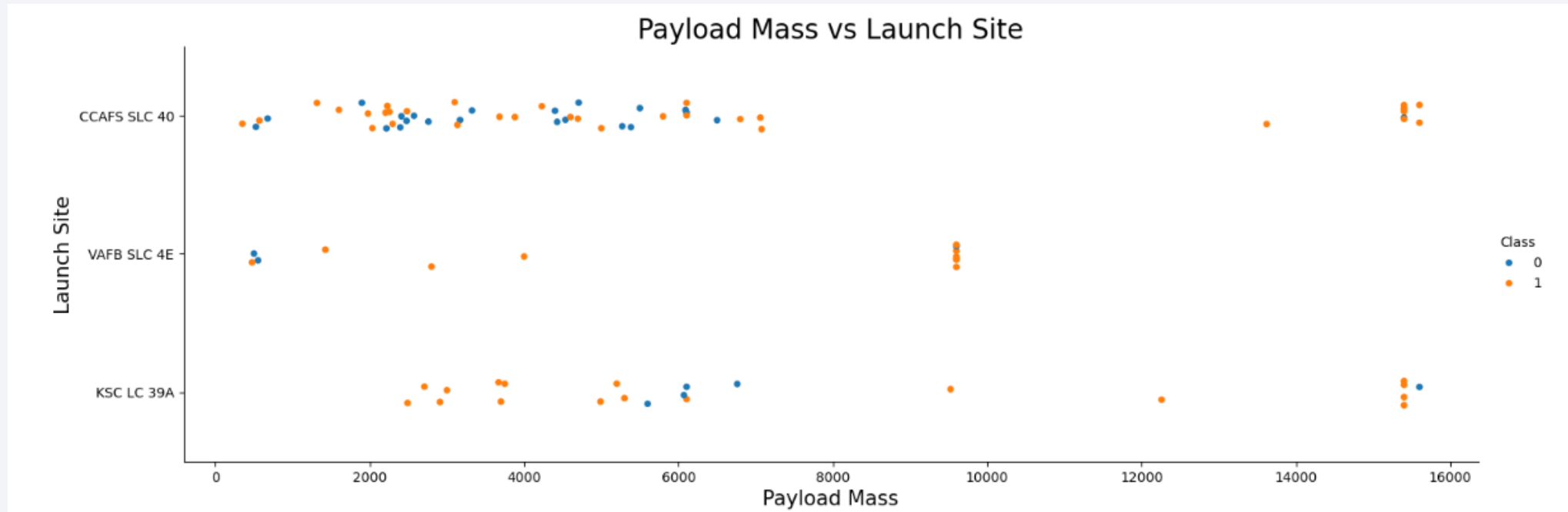
Insights drawn from EDA

Flight Number vs. Launch Site



- **Key Findings**
- Flight number generally increased over time across all launch sites.
- Launches became more frequent as the Falcon 9 program expanded.
- Launch Complex 39A and CCAFS SLC-40 handled many of the later missions.
- This suggests SpaceX relied more heavily on these launch sites as the program matured.

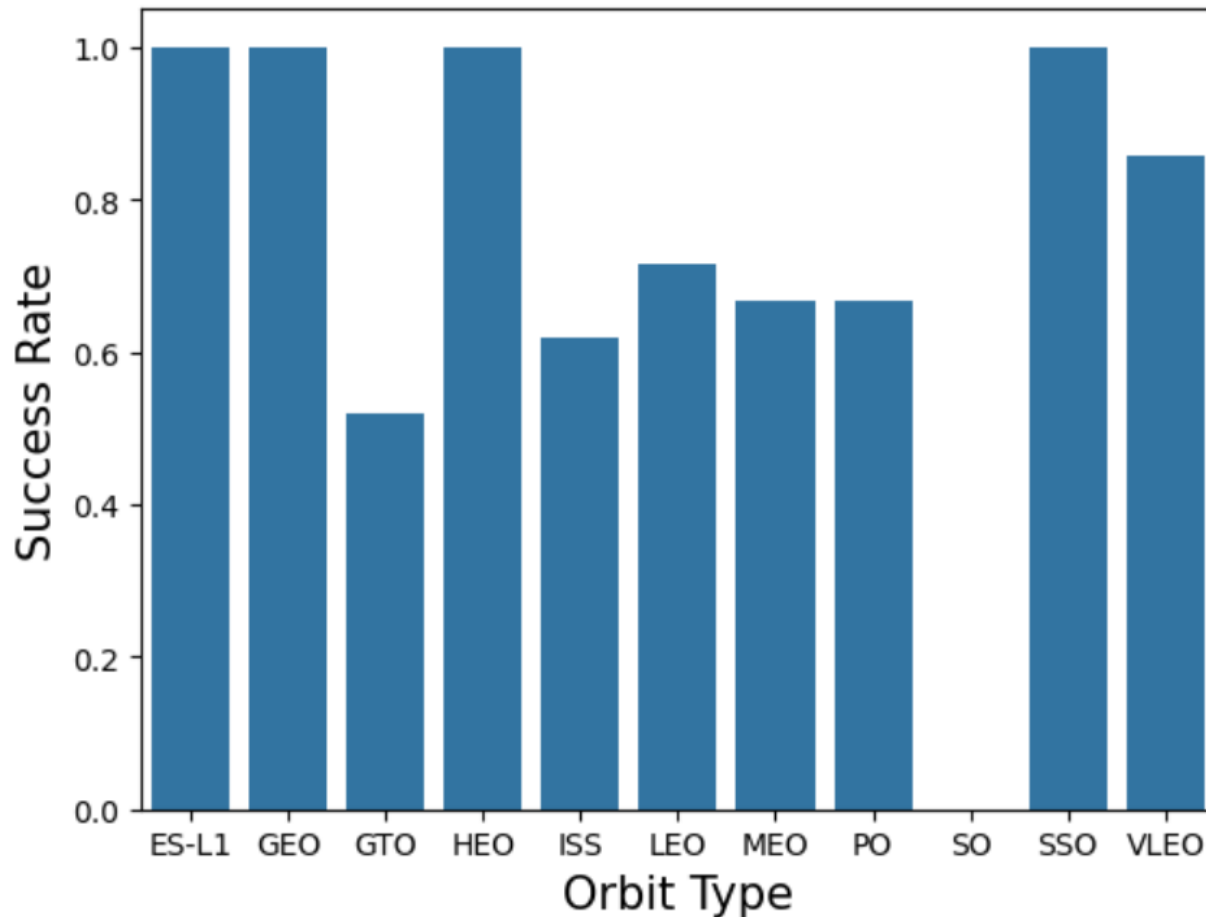
Payload vs. Launch Site



Observations

- Payload mass varies across all launch sites.
- Successful landings occur across a wide range of payload masses.
- KSC LC-39A and CCAFS SLC-40 show many successful launches with heavier payloads.
- Payload alone does not determine landing success, but it is an important factor.

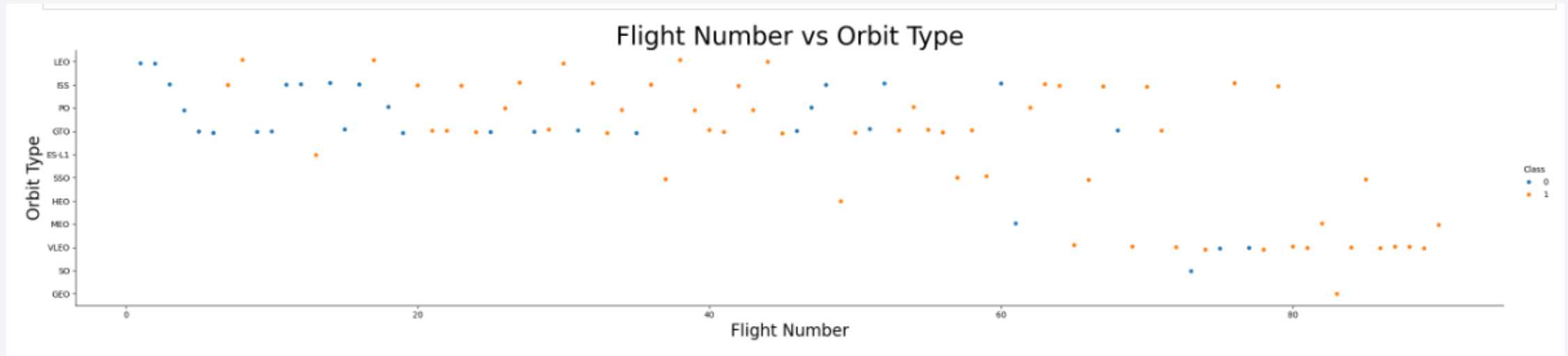
Success Rate vs. Orbit Type



Observations

- Landing success rates vary across orbit types.
- ES-L1, GEO, HEO, and SSO achieved the highest observed success rates.
- GTO had the lowest success rate among the displayed orbit types and SO is practically non-existent.
- Orbit type is a useful feature for predicting first-stage landing outcomes.

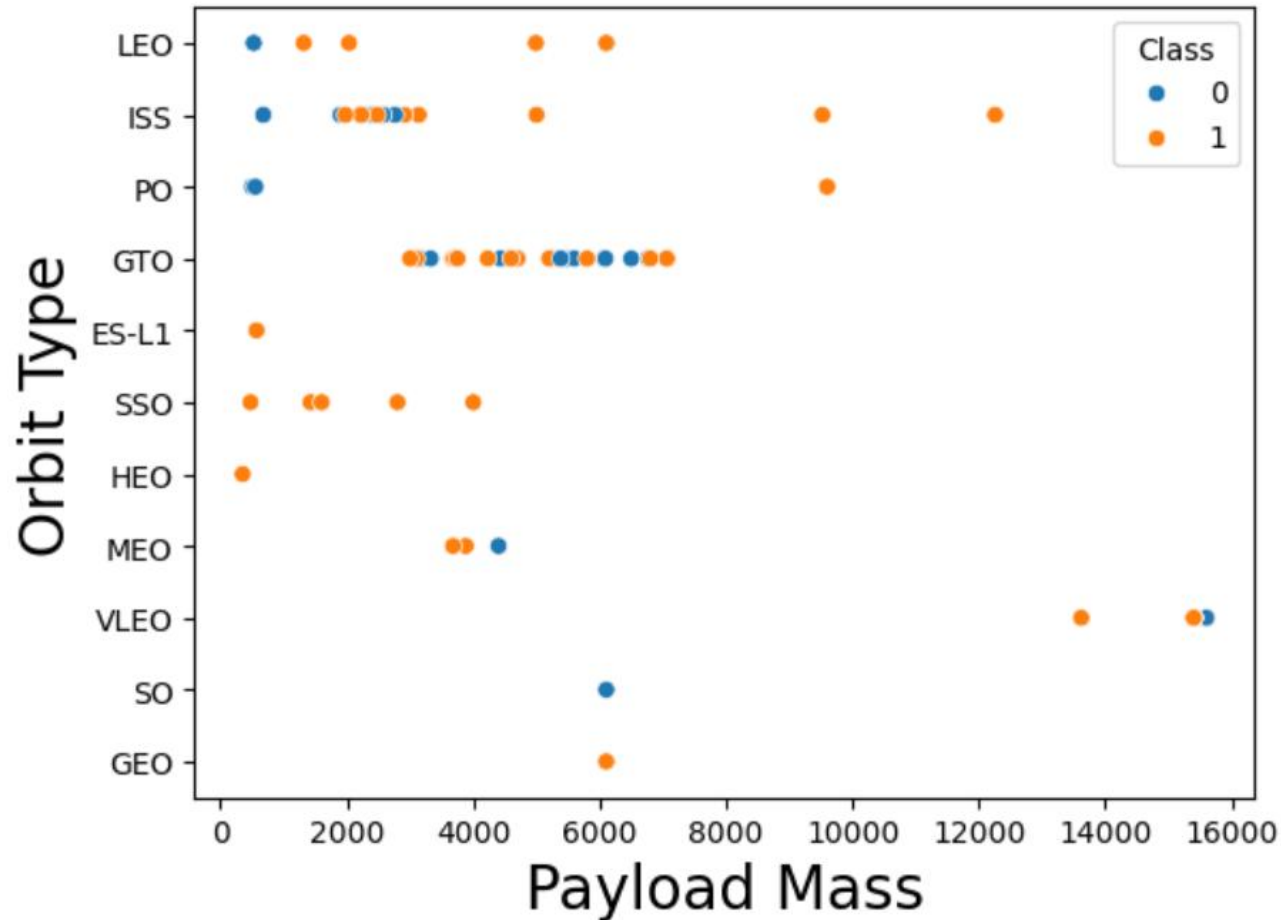
Flight Number vs. Orbit Type



- **Observations**

- Early flights show more unsuccessful landings (blue points).
- Successful landings (orange points) become more frequent in later flight numbers.
- LEO and ISS missions account for many of the successful later launches.
- Flight number and orbit type both appear to influence landing success.

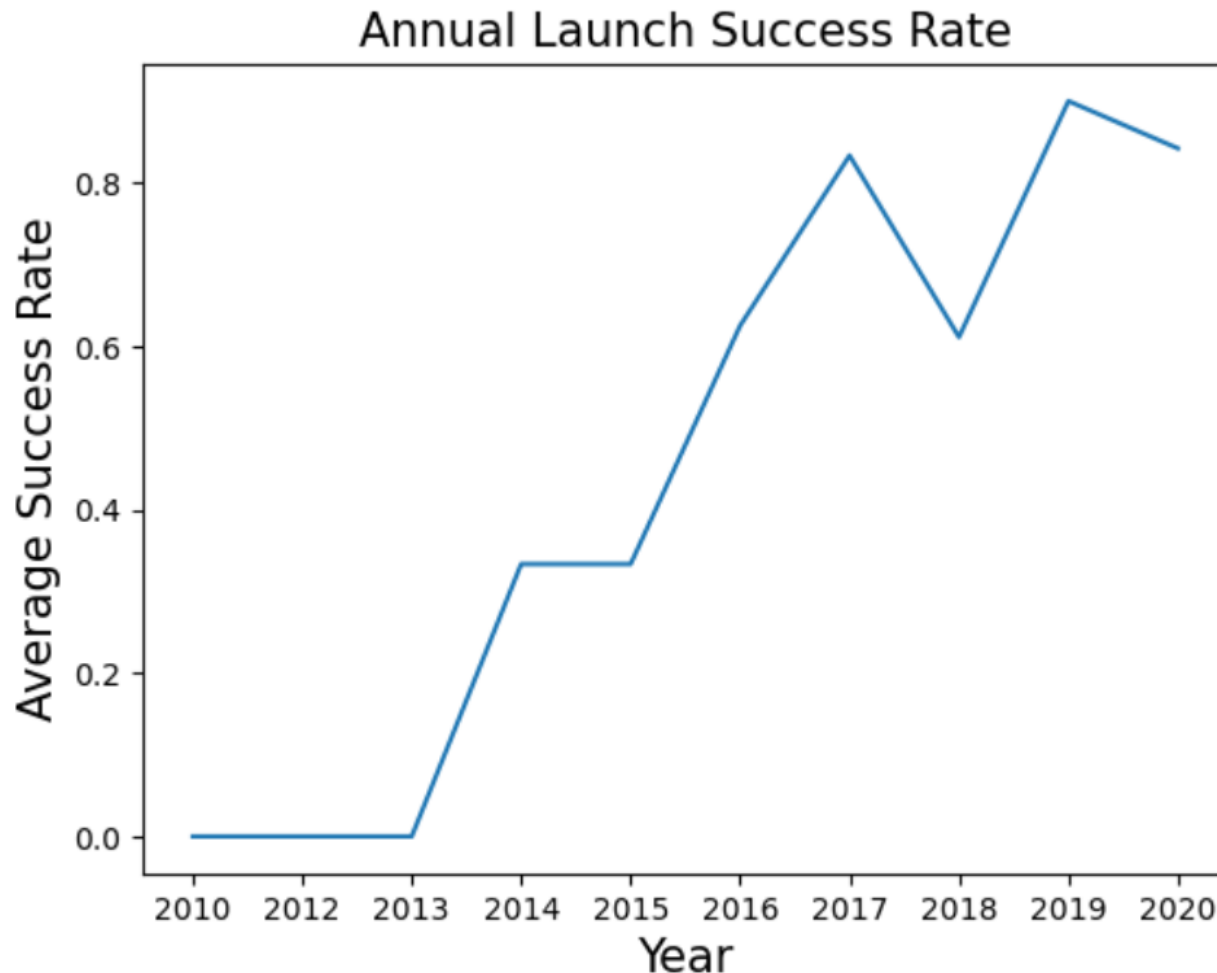
Payload vs. Orbit Type



Observations

- Lower payload missions generally have a higher success rate.
- LEO and ISS missions include many successful launches across different payload masses.
- Higher payloads can still land successfully, but results vary by orbit type.
- Both payload mass and orbit type are useful features for predicting landing outcomes.

Launch Success Yearly Trend



Observations

- Landing success increased year after year.
- Early launches had more failed landings.
- Later missions achieved much higher success rates.
- The results show SpaceX improved its landing technology over time.

All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
In [10]: %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTBL;
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[10]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Observation

- Four unique launch sites were identified in the dataset.
- Most Falcon 9 missions launched from Cape Canaveral and Kennedy Space Center.
- These launch sites were used throughout the analysis to compare launch activity and landing success.

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Query Result

- Retrieved the first five launch records where the launch site name starts with "**CCA**".
- All returned records were from **CCAFS SLC-40**.
- This query demonstrates how the **LIKE** operator can filter text values based on a pattern.
- Pattern matching is useful for searching and grouping related records in a database.

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass_KG
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
Done.
```

Total_Payload_Mass_KG

45596

Query Result

- Calculated the total payload mass carried by **NASA (CRS)** missions.
- Used the **SUM()** function to add the payload mass across all matching launches.
- The result shows the total amount of cargo delivered for NASA missions in the dataset.
- Aggregate functions like **SUM()** are useful for summarizing numerical data.

Average Payload Mass by F9 v1.1

Query Result

- Calculated the average payload mass for missions using the **Falcon 9 v1.1** booster.
- Used the **AVG()** function to find the mean payload across all matching launches.
- The result represents the typical payload carried by the Falcon 9 v1.1 configuration.
- This query shows how aggregate functions can summarize mission performance.

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%%sql
SELECT AVG(PAYLOAD_MASS_KG_)
FROM SPACEXTABLE
WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
Done.
```

```
AVG(PAYLOAD_MASS_KG_)
```

```
2928.4
```

First Successful Ground Landing Date

Query Result

- Retrieved the date of the first successful landing on a ground landing pad.
- Used SQL to filter for successful **Ground Pad** landings and sort by launch date.
- The result identifies the milestone when SpaceX achieved its first successful ground recovery.
- This marked an important step toward reusable rocket technology.

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%%sql
SELECT MIN(Date)
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
Done.
```

MIN(Date)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Query Result

- Retrieved booster versions that successfully landed on a drone ship with payloads between **4,000 kg and 6,000 kg**.
- Four boosters matched the specified conditions:
 - F9 FT B1022
 - F9 FT B1026
 - F9 FT B1021.2
 - F9 FT B1031.2
- This query combines multiple conditions using LIKE and AND to filter the dataset.
- The results show which Falcon 9 boosters successfully completed medium-payload drone ship landings.

```
%%sql
SELECT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome LIKE "Success%"
AND Landing_Outcome LIKE "%(drone ship)"
AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

```
* sqlite:///my_data1.db
Done.
```

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful Mission Outcomes

Query Result

- The SQL query summarizes mission outcomes recorded in the dataset.
- The displayed results show that **98 successful mission outcomes** were identified.
- Successful missions make up the majority of the recorded Falcon 9 launches.
- This result highlights SpaceX's high overall mission success during the analyzed period.

Task 7

List the total number of successful and failure mission outcomes

```
%%sql
SELECT Mission_Outcome, COUNT(*)
FROM SPACEXTABLE
WHERE "Mission_Outcome" IN ('Success', 'Failure')
GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
Done.
```

Mission_Outcome	COUNT(*)
Success	98

Boosters Carried Maximum Payload

Query Result

- Identified all boosters that carried the **maximum payload mass of 15,600 kg**.
- Multiple booster versions achieved this maximum payload capacity.
- The query used `MAX(PAYLOAD_MASS__KG_)` to find the highest payload and returned every booster matching that value.
- This highlights the boosters used for SpaceX's heaviest payload missions.

```
%%sql
SELECT Booster_Version, PAYLOAD_MASS__KG_
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_)
    FROM SPACEXTABLE
);
```

* sqlite:///my_data1.db

Done.

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

```

SELECT CASE
WHEN substr("Date", 6, 2) = '01' THEN 'January'
WHEN substr("Date", 6, 2) = '02' THEN 'February'
WHEN substr("Date", 6, 2) = '03' THEN 'March'
WHEN substr("Date", 6, 2) = '04' THEN 'April'
WHEN substr("Date", 6, 2) = '05' THEN 'May'
WHEN substr("Date", 6, 2) = '06' THEN 'June'
WHEN substr("Date", 6, 2) = '07' THEN 'July'
WHEN substr("Date", 6, 2) = '08' THEN 'August'
WHEN substr("Date", 6, 2) = '09' THEN 'September'
WHEN substr("Date", 6, 2) = '10' THEN 'October'
WHEN substr("Date", 6, 2) = '11' THEN 'November'
WHEN substr("Date", 6, 2) = '12' THEN 'December'
ELSE 'Unknown'
END AS "Month_Name", "Mission_Outcome", "Booster_Version", "Launch_Site"
FROM SPACEXTABLE
WHERE substr("Date", 0, 5) = '2015';

```

* sqlite:///my_data1.db
Done.

Month_Name	Mission_Outcome	Booster_Version	Launch_Site
January	Success	F9 v1.1 B1012	CCAFS LC-40
February	Success	F9 v1.1 B1013	CCAFS LC-40
March	Success	F9 v1.1 B1014	CCAFS LC-40
April	Success	F9 v1.1 B1015	CCAFS LC-40
April	Success	F9 v1.1 B1016	CCAFS LC-40
June	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40
December	Success	F9 FT B1019	CCAFS LC-40

2015 Launch Records

- **Query Result**
- Retrieved all SpaceX launch records from **2015** and displayed the launch month, mission outcome, booster version, and launch site.
- The results show that **most launches in 2015 were successful**, with only **one mission failure (in flight)** occurring in **June**.
- All launches in 2015 were conducted from **CCAFS LC-40**, indicating it was the primary launch site during that year.
- The query used a CASE statement to convert numeric months into readable month names, making the output easier to interpret.

```
%%sql
SELECT "Landing_Outcome", COUNT(*) AS "Count"
FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY COUNT(*) DESC;
```

* sqlite:///my_data1.db
Done.

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query Result

- No attempt was the most frequent outcome, appearing 10 times.
- Successful and failed drone-ship landings were tied, with 5 occurrences each.
- Successful ground-pad landings and controlled ocean landings each occurred 3 times.
- The results show how SpaceX's recovery methods and outcomes changed during the selected period.

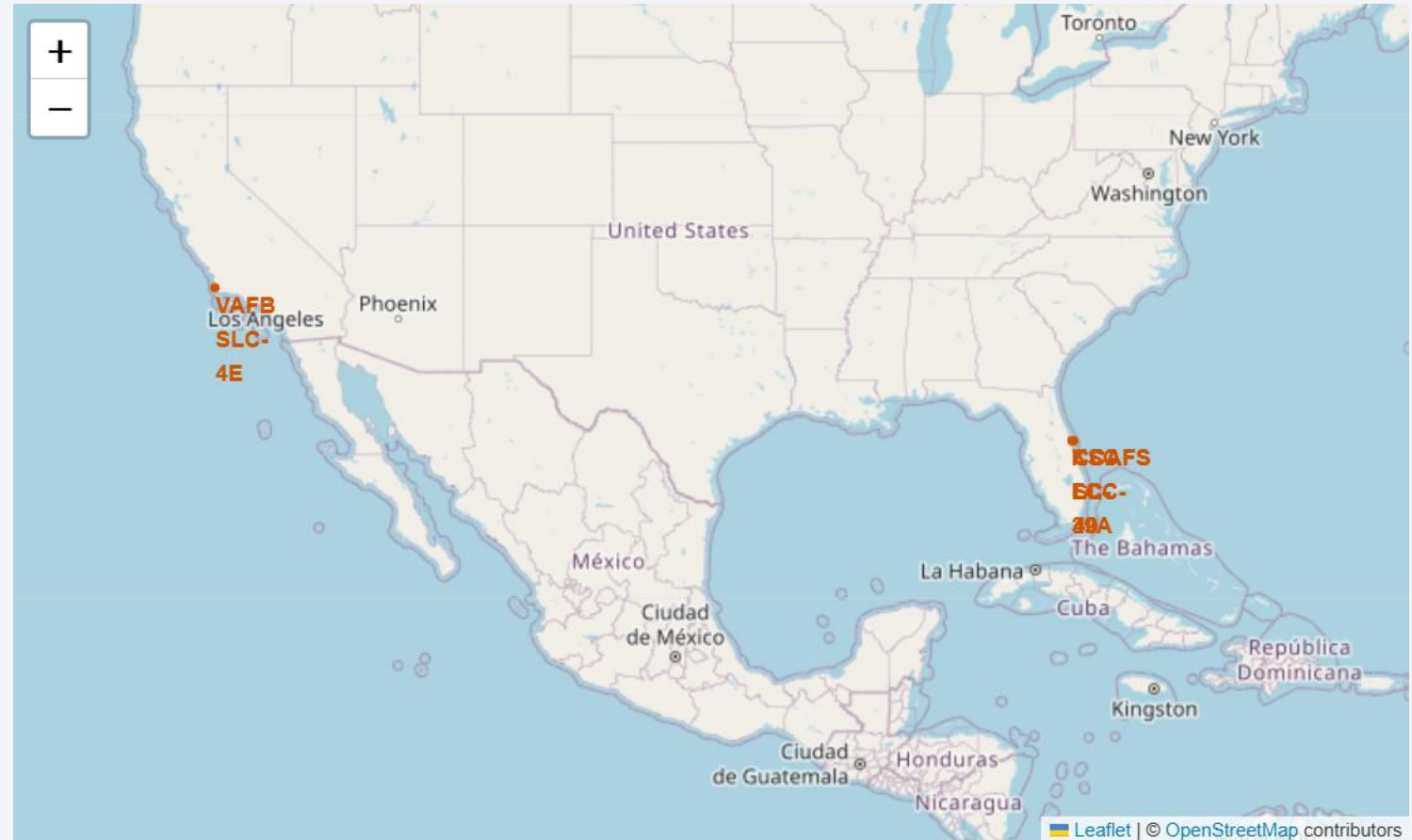
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

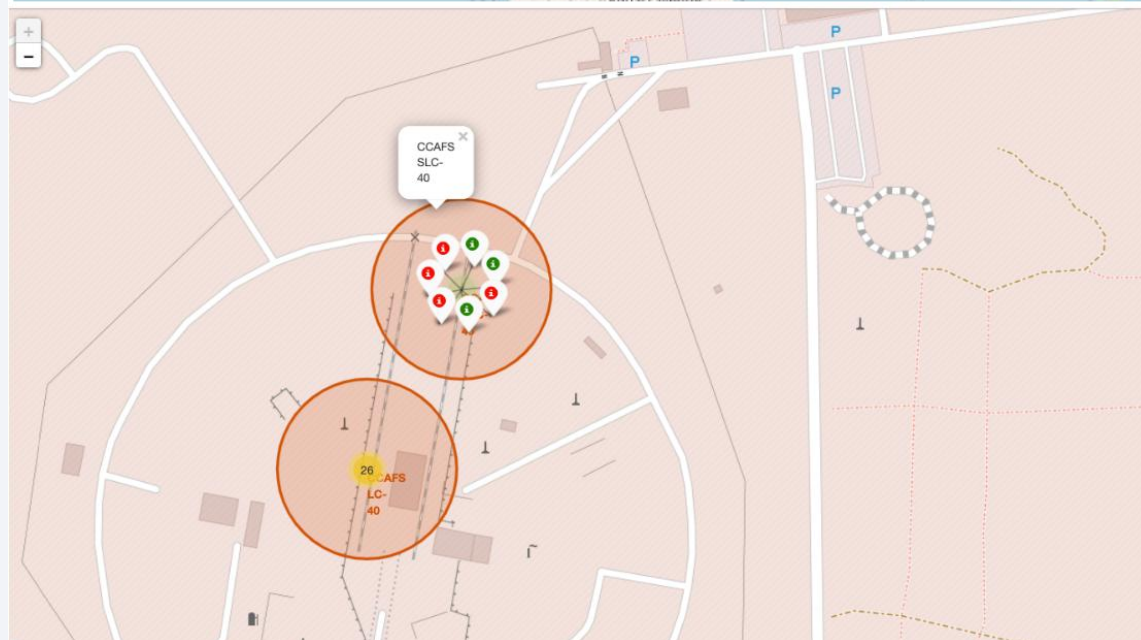
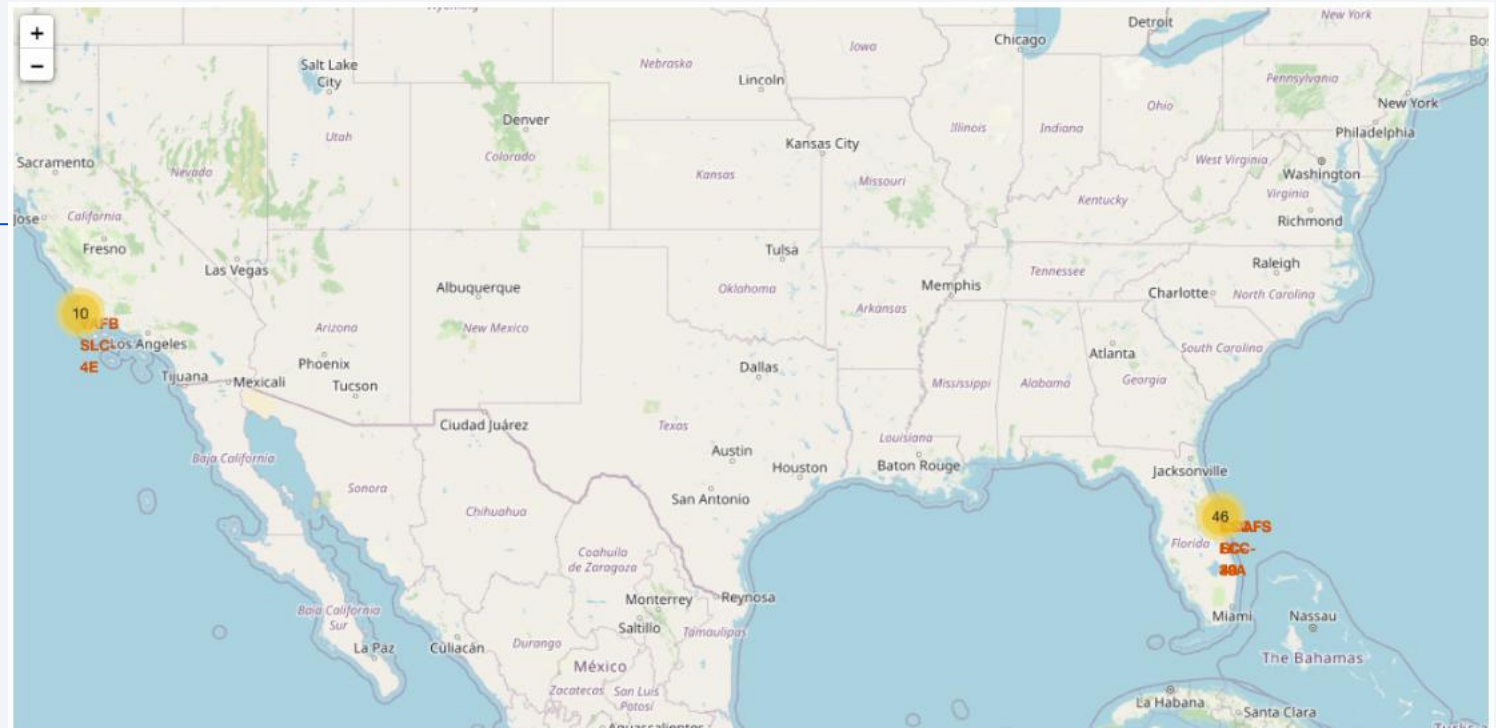
Launch Site Locations

- The Folium map displays the locations of all SpaceX launch sites using interactive markers.
- Most launch sites are located on the **east coast of the United States (Florida)**, with one launch site located on the **west coast (California)**.
- The launch sites are positioned near coastlines, which provide safe launch paths over the ocean and reduce risks to populated areas.
- This map provides a clear geographic overview of where SpaceX launches have been conducted.



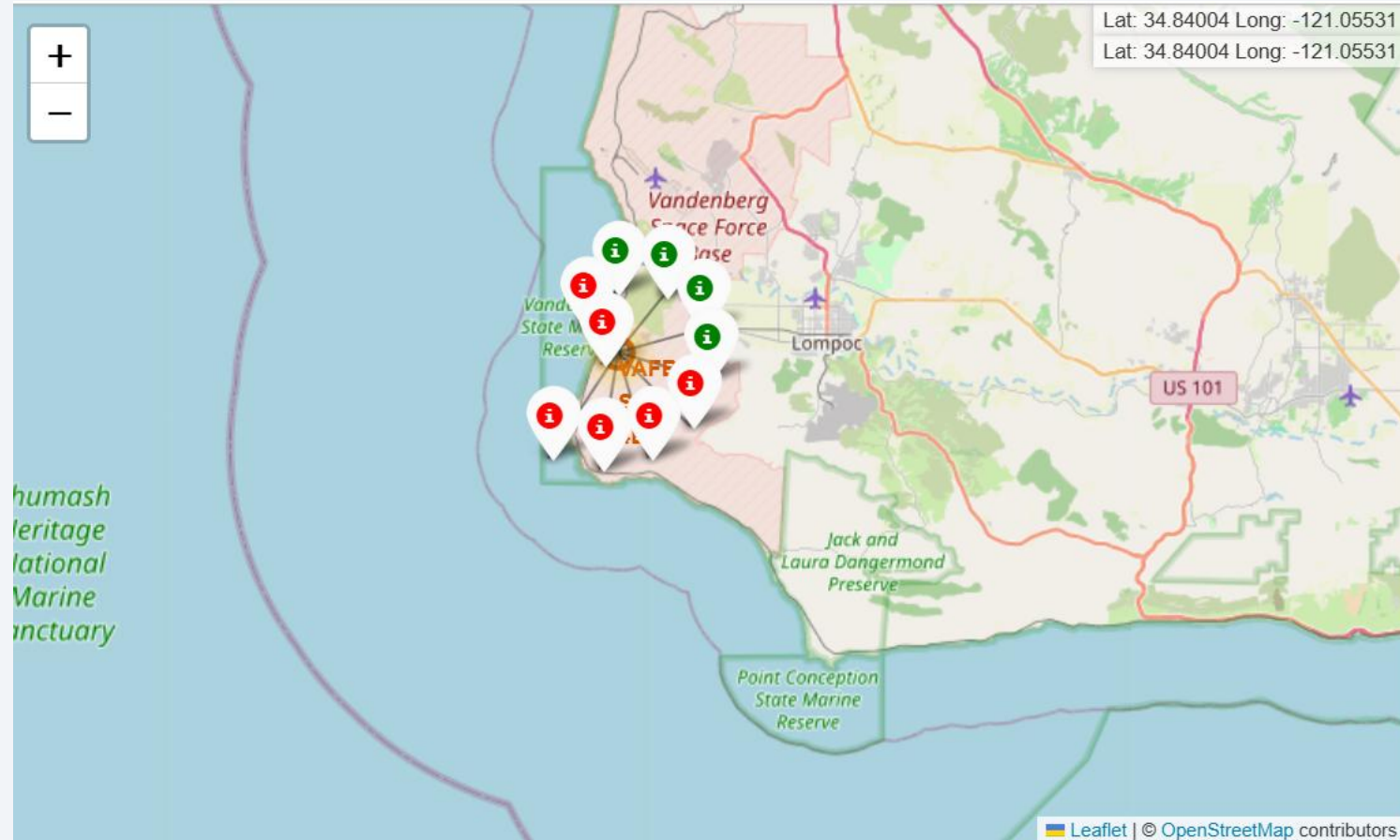
Launch Outcomes by Location

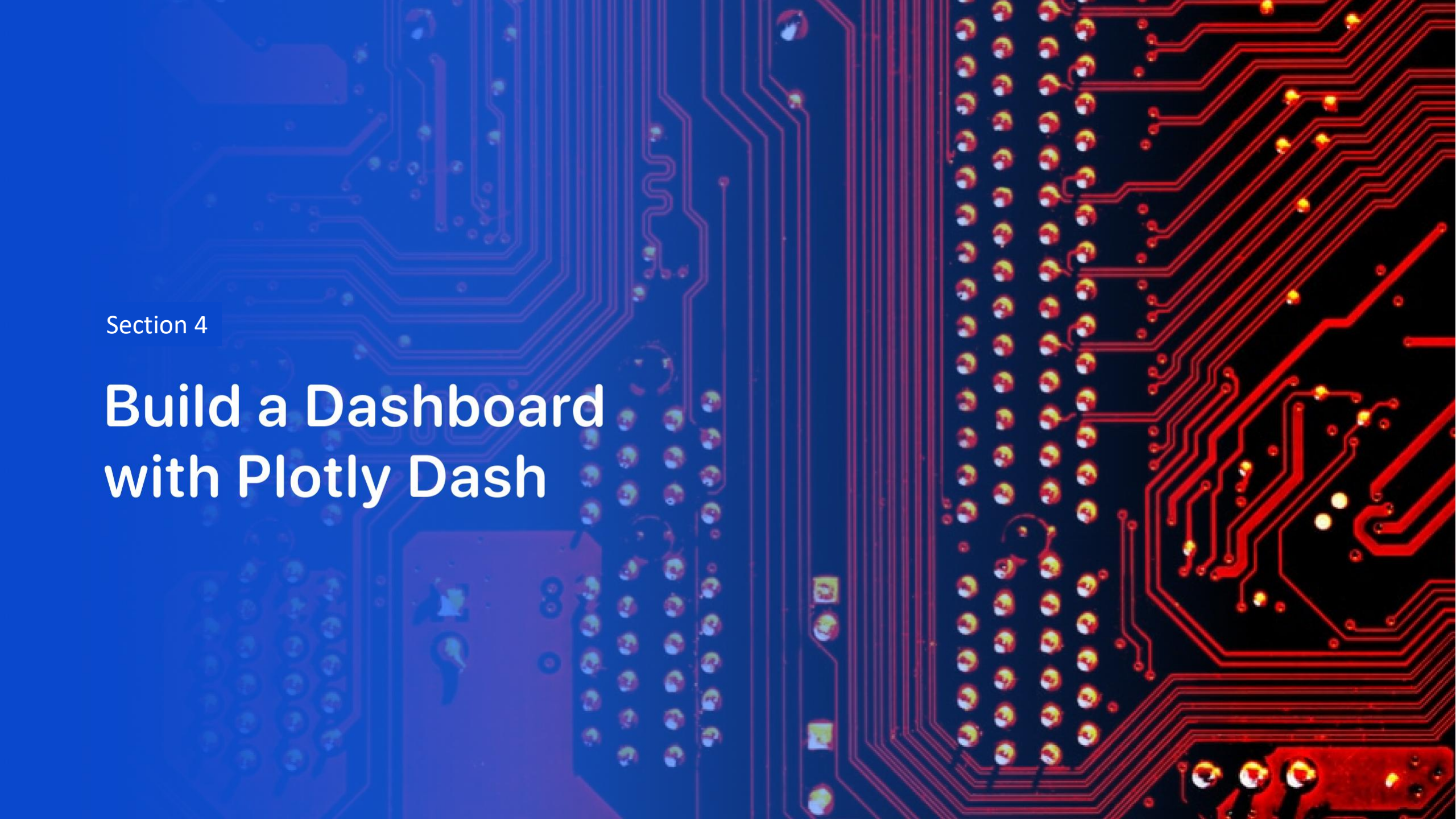
- The map uses **green markers** to represent successful launches and **red markers** to represent failed launches.
- Most launch attempts shown on the map were **successful**, with fewer failed missions.
- The color-coded markers make it easy to compare launch outcomes across different launch sites.
- This visualization provides a quick geographic overview of SpaceX's launch performance.



Launch Site Geographic Overview

- The map displays the selected launch site's geographic location.
- Successful (green) and unsuccessful (red) launches are visualized around the launch site.
- Nearby roads and surrounding geographic features provide context for the launch location.
- The visualization helps compare launch activity while showing the site's surrounding environment.

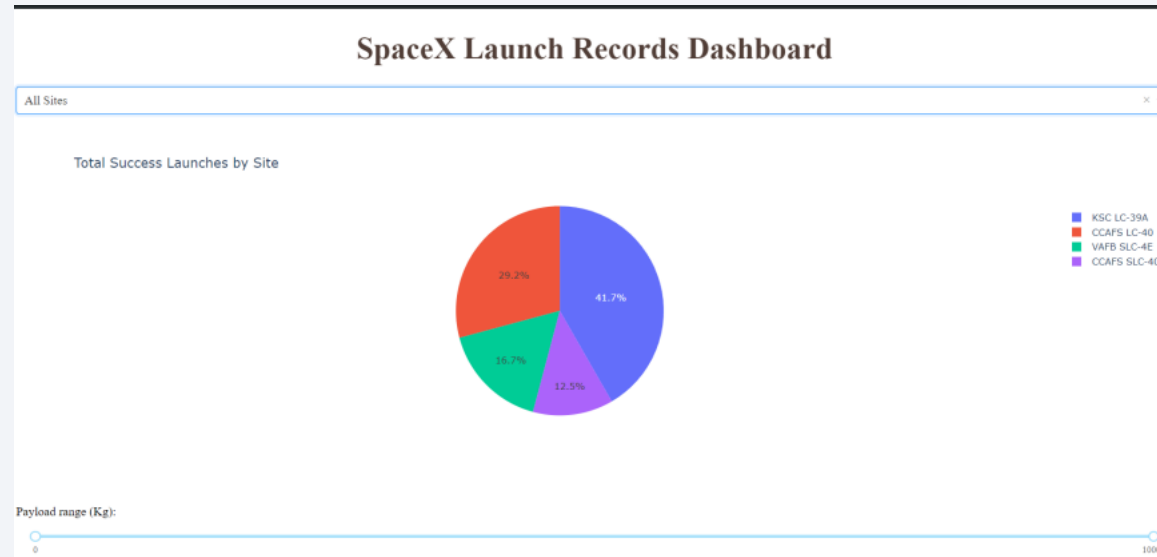




Section 4

Build a Dashboard with Plotly Dash

Launch Success Count by Launch Site



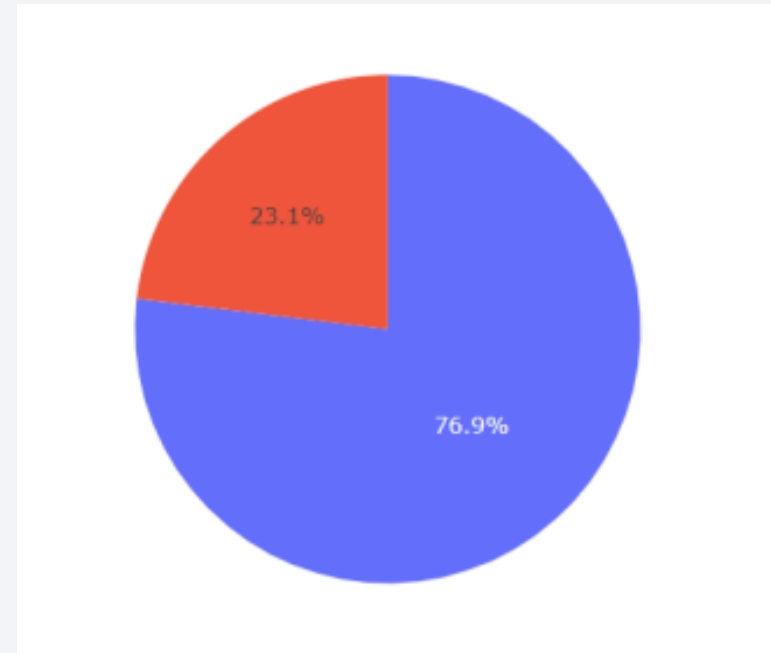
Key Findings:

- **KSC LC-39A** has the highest proportion of successful launches.
- **CCAFS LC-40** is the second-highest share of successful launches
- **VAFB SLC-4E** and **CCAFS SLC-40** account for the remaining successful launches.
- The dashboard allows users to interactively select launch sites and instantly update the visualization.
- The pie chart provides a quick comparison of launch success distribution across all SpaceX launch sites.

Launch Success Ratio – KSC LC-39A

Key Findings:

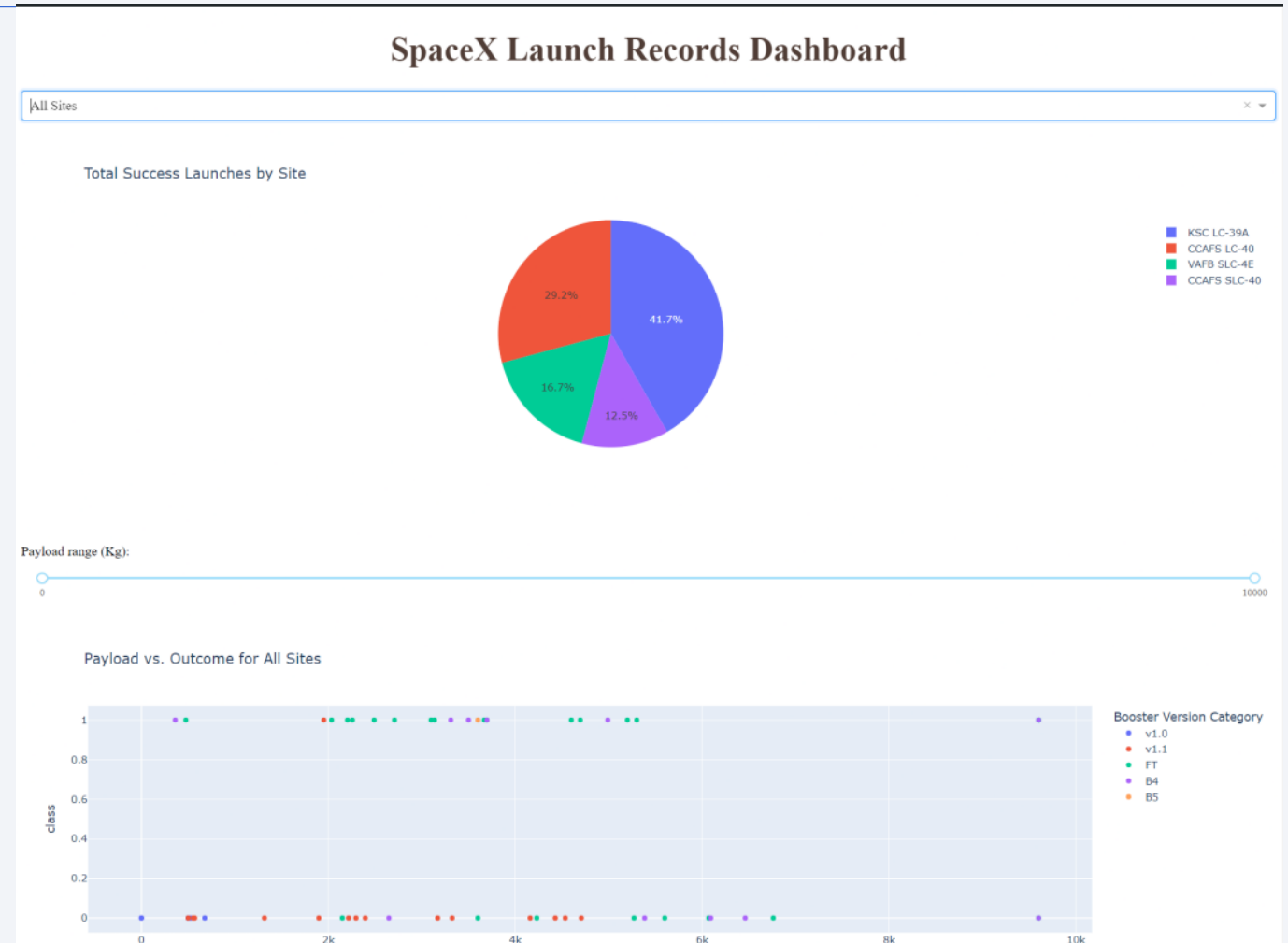
- **KSC LC-39A** has the highest launch success ratio among all SpaceX launch sites.
- **76.9%** of launches from this site were successful, while **23.1%** were unsuccessful.
- The dashboard allows users to filter by launch site and compare success versus failure rates.
- The high success percentage indicates that **KSC LC-39A** is one of SpaceX's most reliable launch sites.



Payload Mass vs. Launch Success Analysis

Key Findings

- The scatter plot shows the relationship between **payload mass** and **launch success** for all launch sites.
- Successful launches are observed across a wide range of payload masses, indicating that **payload alone does not determine mission success**.
- The payload slider allows users to filter missions and analyze how launch outcomes change for different payload ranges.
- Different booster versions appear throughout the graph, suggesting that **multiple booster types successfully handle both light and heavy payloads**.

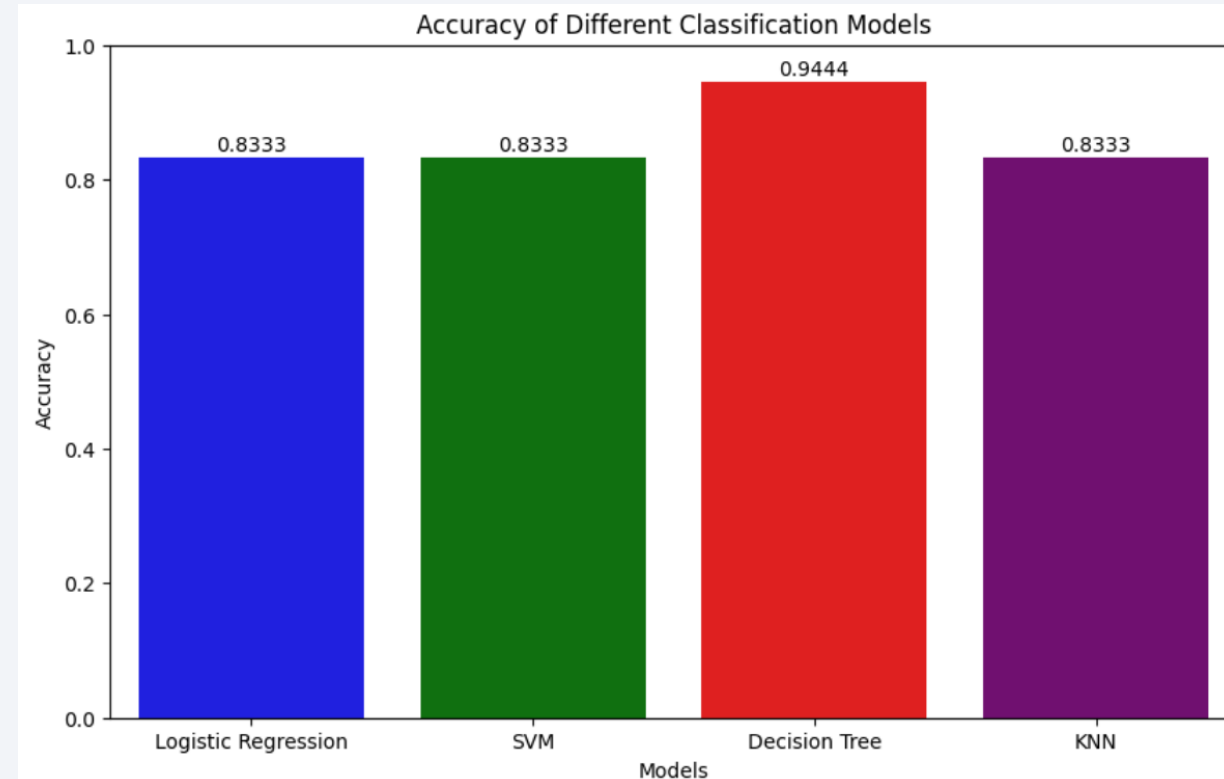


Section 5

Predictive Analysis (Classification)

Classification Accuracy

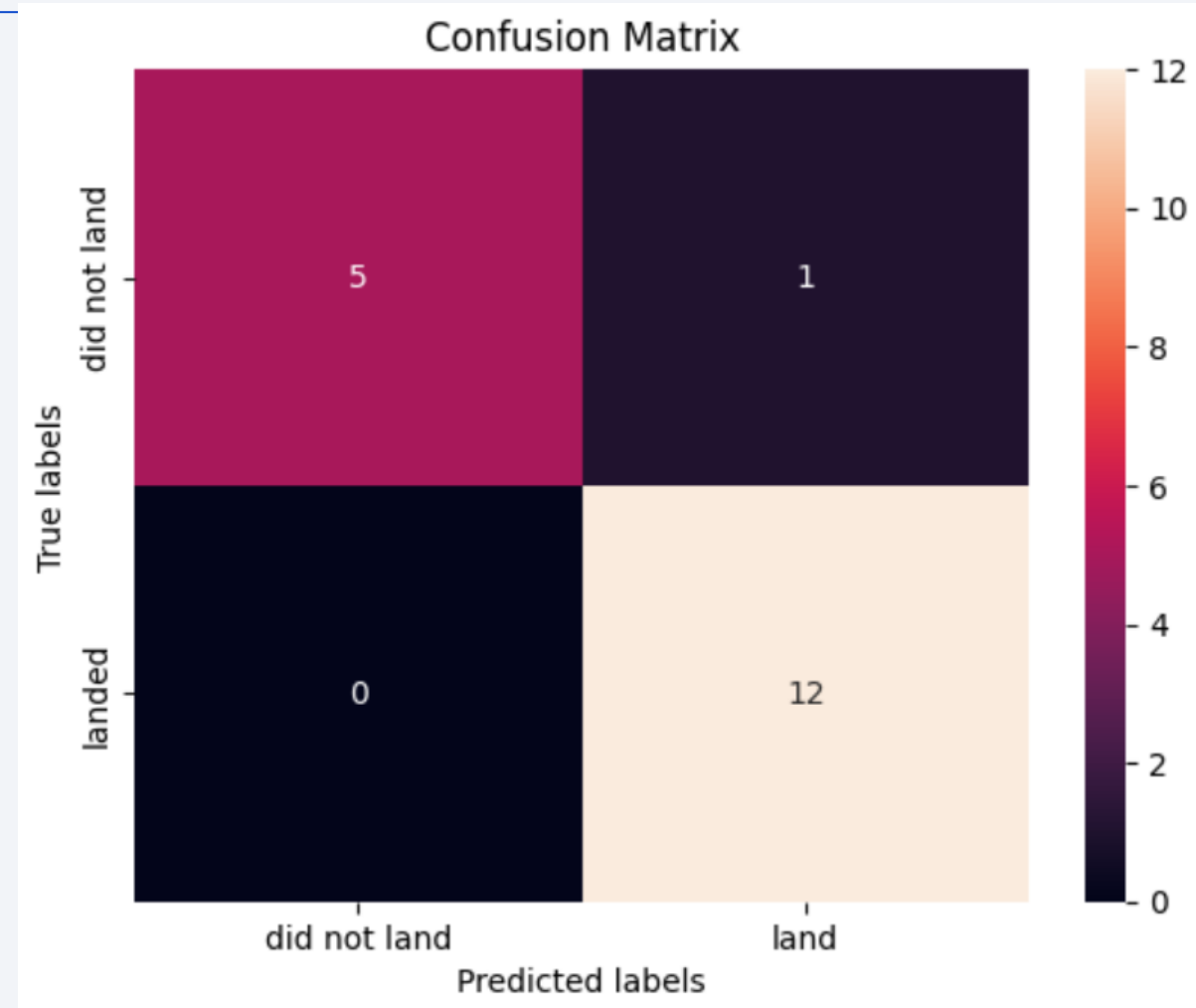
- **Key Findings**
- Four classification algorithms were trained and evaluated: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- Hyperparameter tuning using **GridSearchCV** was performed to optimize each model before evaluation.
- **Decision Tree achieved the highest classification accuracy (94.44%)**, outperforming the other three models.
- Logistic Regression, SVM, and KNN each achieved an accuracy of **83.33%**.
- Based on the evaluation results, the **Decision Tree model was selected as the best-performing classifier** for predicting Falcon 9 landing success.



Confusion Matrix

Key Findings:

- The confusion matrix evaluates the performance of the **Decision Tree classifier**, the best-performing model with **94.44% accuracy**.
- Most test samples were classified correctly, indicating strong predictive performance.
- The matrix contains a high number of **true positives** and **true negatives**, with only a few misclassified observations.
- The small number of **false positives** and **false negatives** suggests the model can reliably predict whether a Falcon 9 booster will successfully land.
- These results confirm that the Decision Tree model is the most suitable classifier for this dataset.



Conclusions

- Successfully collected, cleaned, and explored the **SpaceX Falcon 9 launch dataset** using Python, SQL, and data visualization techniques.
- Identified important factors affecting landing success, including **launch site, orbit type, payload mass, and booster version**.
- Built interactive **Folium maps** and a **Plotly Dash dashboard** to visualize launch locations, mission outcomes, and payload relationships.
- Trained and evaluated four machine learning classification models to predict Falcon 9 landing success.
- **Decision Tree** achieved the highest classification accuracy (**94.44%**) and was selected as the best-performing model.
- This project demonstrates how **data analytics, visualization, and machine learning** can be combined to support data-driven decision making and predict reusable rocket landing outcomes.

Appendix

- GitHub Repository: <https://github.com/raiyan-xy/SpaceX-Falcon9-Landing-Prediction>
- Python Libraries Used:
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Plotly Dash
 - Folium
 - Scikit-learn
- SQL Database: SQLite
- Machine Learning Models:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Tree
 - K-Nearest Neighbors (KNN)

Thank you!

